3D Designer: Step-by-Step Usage Flow

This guide walks through the full workflow, from installing the app to producing a BOM, drawings, labels and CNC files.

Overview

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PHASE 1  Install
PHASE 2  One-time setup (Items → Material Map → Rules → Library → Settings)
PHASE 3  Create a Design Project
PHASE 4  Get parts in (design in browser OR upload file)
PHASE 5  Edit and review the model
PHASE 6  Save (auto: joints → holes → nesting → BOM job)
PHASE 7  Review the BOM
PHASE 8  Outputs (drawing, cutting plan, labels, CNC, render)
PHASE 9  Revise (change parts → save → new BOM version)
```

Roles: Phases 1 and 2 are done once by an admin. Phases 3 to 9 are done by designers for each job.

Phase 1: Install (admin, once)

Step	Action	Result
1.1	`cd ~/frappe-bench`	
1.2	`bench get-app <repo-url> --branch main`	App downloaded
1.3	`bench build --app frappe_3d_designer`	Three.js bundle built
1.4	`bench --site <site> install-app frappe_3d_designer`	DocTypes, custom field `BOM.design_project`, 4 starter connection rules, 5 starter Operations with workstations (Panel Cutting, Edge Banding, Drilling, Assembly, Packing) and 10 starter hardware Items and library components (Screw, Handle, Hinge, Cam Lock and so on)
1.5	`bench --site <site> migrate` then `bench restart`	App live

Requires Frappe/ERPNext v16, Python 3.14+, Node 24.

Phase 2: One-time setup (admin)

Do these in order, because each step depends on the one before.

Step 2.1: Raw material Items

Create Items in ERPNext:

- Boards: stock UOM `Nos` (sheet basis) or `Square Meter` (area basis).
- Solid or volume materials: `Cubic Meter`.
- Edge banding: `Meter`.
- Hardware (hinges, handles, dowels, cams, screws): `Nos`.

Step 2.2: Design Material Map

Create one record per board or material.

Field	What to enter
Raw material item	The Item from 2.1
Calculation basis	Sheet / Area / Volume / Length / Nos
Sheet size, kerf, trim	Only for Sheet basis (used by nesting)
Has Grain	Tick if the material has a grain direction
Wastage %	Added to the BOM quantity (not for Sheet)
Colour	Used in 3D and renders
Edge banding item	Item used for banded edges
Aliases	One per line, e.g. `WHITE_18`, `MDF18`

Step 2.3: Design Connection Rules

Open the 4 starter rules (Dowel 8x35, Cam and Dowel, Back Panel Screw, Back Panel Groove). For each:

- Set the max spacing, min connectors per joint and offset from the ends.
- Set the face and edge hole diameters and depths.
- Add hardware items per connector (Design Connection Item rows).
- Set the operation and minutes per connector.

Step 2.4: Design Library Components

Install already adds starter hardware (Screw, Gum, Handle, Hinge, Cam Lock, Wooden Dowel, Furniture Leg, Drawer Runner, Edge Band Tape, Nails) as Items and library components. Add your own hinges, handles, legs, runners and profiles here. They appear in the designer's **Add part** list, grouped by category.

Step 2.5: Designer Settings

- Default connection rule and thin-panel rule, plus the thin-panel limit (for back panels).
- Company and finished-good item group.
- Fallback material, used when a part's material is not found.
- BOM options: make it default, submit automatically.
- Unit of uploaded models (GLB is usually metres).
- Detect Joints Automatically (tick to enable).
- Operations table (workstation and minutes per operation). Leave it empty to use the starter routing (Panel Cutting, Edge Banding, Drilling, Assembly, Packing). Untick *With Operations* to produce a BOM without a routing.

Setup is complete when: every material you will use has a Material Map, and the rules point to real hardware Items.

Phase 3: Create a Design Project (designer)

Step	Action
3.1	Go to **Design Project** → **New**.
3.2	Enter the title and customer or reference.
3.3	Save. The form shows the parts table, a 3D preview and the BOM link.

Step	Action
3.4	Click **Open in 3D Designer**, or go to `/app/three-d-designer`.

Phase 4: Get parts in (choose A or B)

Route A: Design in the browser

Step	Action
A.1	For a cabinet, click **Cabinet**. Enter the size, plinth, shelves, doors and back type (overlay or grooved).
A.1b	For other furniture, click **Furniture** and pick the type (see table below).
A.2	Choose the material for carcass, doors and back.
A.3	Confirm (**Build** for furniture). The parts are generated.
A.4	Add extra panels from **Add part** → **Panels**, or hardware (screws, glue, handles, hinges and so on) from the other groups in the same list.

Furniture types and their options:

Type	Options
Dining Table, Coffee Table, Study Desk, Bench	Width, depth, height, top thickness, leg size, leg style (square or round), apron
Chair	Same, plus seat height and number of back slats
Bed	Width, depth, headboard height, leg/post size, number of slats (about 14)
Bookshelf, TV Unit	Width, depth, height, board thickness, shelves (built as a cabinet)

Choosing a type fills in typical sizes, which you can change. Tick **Replace current parts** to start over, otherwise the new piece is placed beside the existing parts.

Route B: Upload a file

Step	Action
B.1	Prepare the file (see formats below).
B.2	Attach it in the **Design File** field on the Design Project.
B.3	Save. The parts are read into the parts table.

Format	Preparation
GLB / GLTF	Single embedded file. One mesh = one part. Name meshes `Part name__Material`.
OBJ	Same naming rule.
CSV / XLSX	Header row within the first 10 rows. Columns: Part, Material, Length, Width, Thickness, Qty, Edge long, Edge short, Item code. German headers also work.
JSON	Download from the 3D Designer menu.

PYTHA, SketchUp, Blender and Fusion native files are not supported. Export GLB, OBJ or a parts list from them.

Check: if the form shows "unmapped material", add that name as an alias in a Design Material Map (Step 2.2) and save again.

Phase 5: Edit and review the model

Step 5.1: Edit parts

Select a part and change:

- Material, grain, shape (panel or cylinder)
- Length, width, thickness, qty
- Orientation, exact X/Y/Z position
- Edge banding (long and short edges)
- Hardware item (added per piece). To add a loose item such as a screw or handle, use **Add part** and choose it from the library groups.
- Machining rows: Drill (face, X, Y, diameter, depth), Route, Pocket, Edge Profile, Miter, Groove

Also available: **duplicate**, **rotate 90°**, **delete**, and **drag to move** (5 mm snap).

Step 5.2: Inspect

Tool	Use
3D / Front / Top / Side	Check dimensions from every angle
Section plane	Cut through the model (height, width, depth)
Explode slider	See how the parts fit together
Drilling overlay	Check auto-generated and manual boreholes
Presentation mode	PBR materials, wood grain, soft shadows, for client views

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Phase 6: Save (what happens automatically)

Click **Save design**. Then:

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Save
 ■■ 1. Parts hash checked (changed? continue)
 ■■ 2. Joints detected between touching panels
 ■     ■■ rule chosen: default, or thin-panel rule for back panels
 ■■ 3. Connectors placed along each joint
 ■     ■■ boreholes generated (face and edge holes)
 ■■ 4. Nesting (MaxRects) per sheet material
 ■     ■■ uses kerf, trim, grain → sheet layouts, with a cut list and waste % per sheet
 ■■ 5. Background job: BOM sync (Phase 7)
```

Errors appear at the top of the Design Project form.

Phase 7: Review the BOM

Step	Action
7.1	Wait for the live notification, or click **Generate BOM** to run it now.
7.2	Open the linked BOM from the Design Project.
7.3	Check the rows.

How the BOM is built:

- Sheet materials: whole sheets from the cutting plan. Each sheet layout lists its cut pieces and waste %.
- Cylinders (round legs, screws) cannot be nested, so they are bought by area and still appear in the BOM.
- Area, volume and length materials: calculated from the part sizes, plus wastage %.
- Banded edges: edge banding item, in metres.
- Joints: connector hardware and operation minutes.
- Operations: from Designer Settings, or the starter routing if none is set. Per-piece operations scale with part count, others are fixed minutes.
- Parts with a Hardware Item: added per piece.
- The same raw material across parts is merged into one BOM row.

Behaviour:

- If a draft BOM is linked, it is updated in place.
- If the linked BOM is submitted, a new BOM version is created and the old one stays for existing Work Orders.
- If there is no finished good item, one is created from Designer Settings.

Next in ERPNext: create a Work Order from the BOM as usual.

Phase 8: Outputs

Output	Steps	What you get
Technical drawing	Menu → Technical drawing, or the print format *Design Technical Drawing*	Front, side and top views, overall dimensions, part balloons, title block, Section A-A
Cutting plan	Menu → Cutting plan, or print format *Design Cutting Plan*	Sheet diagrams and part list
Part labels	Print format *Design Part Labels*	100 × 50 mm labels, QR code per piece, grain and edge info
CNC export	Menu → Export for CNC	Zip attached to the design (see below)
Render	Presentation mode → Save render	PNG attached to the design
Model files	Menu → Download GLB / JSON	For sharing or re-import

CNC zip contents:

- `cutlist.csv`: for a panel saw or optimiser
- `program.json`: neutral program per part
- `dxf/<label>.dxf`: one per part, with layers `OUTLINE`, `DRILL_…`, `GROOVE_…`, `ROUTE_…`, `POCKET_…`, `PROFILE_…`, `MITER_…`
- `sheets/*.svg` and `drawing.svg`

Map the DXF layer names to operations once in your CAM software.

Phase 9: Revise a design

Step	Action
9.1	Open the Design Project, then the 3D Designer.
9.2	Change parts, materials or sizes.
9.3	Save. Joints, holes and nesting are recalculated.
9.4	The BOM is updated (draft) or a new version is created (submitted).
9.5	Re-print drawings, labels and the CNC export.

Troubleshooting

Symptom	Cause	Fix
"Unmapped material" error	Name is not in any Material Map	Add the name as an alias, or set a fallback material in Designer Settings
GLB model is the wrong size	Wrong unit	Set the model unit in Designer Settings
No joints or holes	Joint detection is off, or the rule has no connectors	Tick *Detect Joints Automatically*, check the rule values
No hardware in the BOM	Rule has no connection items	Add hardware Items to the Connection Rule
BOM not updated	Background job still running or failed	Check the form error banner and the background jobs list
Edge banding missing in the BOM	No edge banding item on the material	Set it in the Material Map
"Could not be placed on a sheet" error	A board part is larger than the sheet or no layout was produced	Check the part size and the sheet size, kerf and trim in the Material Map
BOM has no operations	*With Operations* is unticked, or Operation masters are missing	Tick it in Designer Settings; saving the BOM re-creates the starter Operations and workstations
Nesting looks wrong	Wrong sheet size, kerf or trim	Fix them in the Material Map

Limits

- Parts are rectangular panels or cylinders, with rotation in 90° steps.
- Nesting is rectangular only.
- Presentation mode is real-time rendering, not ray tracing.
- The DXF export is generic. Machine-native formats (MPR, CIX and so on) need a postprocessor in `cnc_export.py`.